

专业年级: 2020 级数据科学与大数据技术

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1. What is the formal definition of big-O?

2. Let input being 2 n -digit numbers x and y , and output being product $x*y$, please write down a recursive algorithm that needs only three recursive multiplications (i.e. the Karatsuba Multiplication).

3. Please write down the Pseudocode for the Partition step in Quick Sort.

4. Please write down the Random Contraction Algorithm by Karger.

5. What's the running time of the Dijkstra's algorithm with Heap? Please also explain why.

6. In the problem of finding max-weight independent set of a path graph G , what is the relationship between the solution of G and G' , where G' is G with the last node being deleted. Also, what is the relationship between the solution of G and G'' , where G'' is G with the last two nodes being deleted. Do explain your reasoning.

7. Please write down the non-recursive algorithm to solve the Knapsack problem.

8. Please explain how do we use heap to change the original selection sort algorithm of $O(n^2)$ to a new algorithm, which is as fast as a merge sort algorithm.

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二、证明题 （每小题 12 分，共 36 分）

1. Please prove that if $T(n) = a_k n^k + \dots + a_1 n + a_0$, then $T(n) = O(n^k)$.

2. Please prove the correctness of the Dijkstra's algorithm. That is to prove that for any directed graph with nonnegative edge lengths, Dijkstra's algorithm correctly computes all shortest path distances.

3. Please prove that in the scheduling problem, if we sort jobs by the ratio of weight over length and then do jobs in decreasing order, we will always get the correct answer.